

## PARTS LIST

## Semiconductors:

|            |                              |
|------------|------------------------------|
| IC1        | - ATmega328 microcontroller  |
| IC2        | - 7812, 12V regulator        |
| IC3        | - 7805, 5V regulator         |
| T1         | - MPS2222 npn transistor     |
| BR1        | - 1A bridge rectifier module |
| LED1, LED2 | - 5mm LED                    |
| D1         | - 1N4007 rectifier diode     |
| ZD1        | - 3.3V zener diode           |
| Bluetooth  | - HC-05 Bluetooth module     |

Resistors (all 1/4-watt,  $\pm 5$  per cent carbon):

|        |               |
|--------|---------------|
| R1     | - 680-ohm     |
| R2     | - 100-ohm     |
| R3     | - 10-kilo-ohm |
| R4, R5 | - 1-kilo-ohm  |

## Capacitors:

|        |                                  |
|--------|----------------------------------|
| C1     | - 1000 $\mu$ F, 35V electrolytic |
| C2, C3 | - 0.1 $\mu$ F ceramic disk       |
| C4     | - 10 $\mu$ F, 16V electrolytic   |
| C5, C6 | - 22pF ceramic disk              |

## Miscellaneous:

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| X1              | - 230V AC primary to 15V, 500mA secondary transformer |
| PZ1             | - Piezobuzzer                                         |
| X <sub>AL</sub> | - 16MHz crystal oscillator                            |
| S1              | - Push-to-on switch                                   |
| S2              | - On/off switch                                       |
| CON1            | - 2-pin, 3.5mm connector for the electric lock        |
| CON2            | - 4-pin berg strip connector for Bluetooth            |
| BATT.1          | - 12V electric strike lock YS 130                     |
|                 | - 12V, 7Ah battery                                    |

project. Use a 28-pin IC base for the microcontroller. After assembling the circuit on the PCB, check proper connections, especially for the power supply. Connect the two leads of the electric strike to CON1 in the PCB. Next, connect the Bluetooth module to the circuit at CON2. Note that the lock and Bluetooth require 12V DC and 3.3V DC power supplies, respectively.

Now copy the code from this month's EFY DVD into your Arduino IDE. Compile it and load into the ATmega328. Remove the ATmega328 from the Arduino board and insert into the IC base in the circuit. Next, switch on the circuit. Make sure that LED1 glows and LED2 is off. Note that

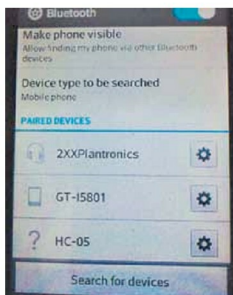


Fig. 8: Pairing mobile with HC-05 Bluetooth

12V battery is used only when mains power fails.

Turn on your Android phone and open the Bluetooth application. Pairing your phone's Bluetooth with another Bluetooth device is relatively simple. Instructions for the same are available in your mobile phone's manual. The steps are just about the same for any Android phone or tablet, and the instructions that came with your handset will fill in the rest of the details.

The first time you use a new Bluetooth device, you need to pair it with your phone so that both devices know how to connect securely to each other. Thereafter, they connect automatically.

To connect to a Bluetooth device, first turn on your phone's Bluetooth. Select 'Device type to be searched' as 'All devices,' then click 'Search for devices' option.

As shown in Fig. 8, you can see the nearby device as 'HC-05' in your Android device. This means that your mobile phone is connected with the

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on [www.efymag.com](http://www.efymag.com) website.

Bluetooth module in the circuit.

Now open the Bluetooth SPP application in your phone, select 'Real-time mode' option.

You can see 'Enter the password' on the mobile screen. Enter '555' as the password, then click 'Send' button. This control signal is received by the Bluetooth and fed to the microcontroller. The microcontroller processes the signal and its pins 15 and 19 go high for a brief period of time. Piezobuzzer PZ1 connected to pin 19 sounds for two seconds, indicating that a control signal has been received and the lock is open. At the same time, pin 15 goes high for 10 seconds, switching on transistor T1. The transistor conducts, enabling the lock to open. The lock closes automatically after 10 seconds. That is, whenever a valid password is received, the electric lock opens for ten seconds and then closes automatically.

If everything is working as mentioned above, you can install the electric strike or electric lock on the door frame and connect the 12V battery. In case you face any problem in the circuit, press reset switch S1 momentarily and try again. If problem persists, you may check the test point voltages as per the table.

**Note.** To extend the life of your mobile phone battery, turn off Bluetooth application when not in use. ●

The author is a B.Tech in electronics and communication from Punjab Technical University, Punjab



For More  
Test and Equipment Stories  
Visit [www.electronicbsb2b.com](http://www.electronicbsb2b.com)